Plasma-Echo-Constrained Kinetic Memory
A Reduced-Order Falsification Framework

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Abstract

This paper studies a reduced two-pulse plasma-echo model and a synthetic delay-embedding benchmark. It distinguishes kinematic rephasing, phenomenological collisional attenuation and synthetic forecasting. It is not a self-consistent Vlasov-Poisson simulation, does not train a neural operator on kinetic data, and does not analyse laboratory or spacecraft observations. No new plasma mechanism or experimental validation is claimed.

1 Question and scope

Can an echo-centred intervention distinguish a reduced state that retains recoverable phase information from one whose echo response is attenuated by a declared model? The second pulse is an intervention. The target is the later echo-mode response under pulse protocols withheld from fitting. The rephasing time is kinematic; attenuation is a model definition; forecasting is synthetic. These layers must not be conflated.

2 Phase mixing

A Fourier perturbation in a freely streaming kinetic system carries the velocity phase exp(-ikvt). Velocity integration can make a low-order moment small while the distribution develops fine velocity-space structure. Cancellation is not microscopic erasure and is not by itself thermodynamic entropy production. Landau damping is self-consistent; free streaming is used only to expose echo geometry.

3 Established echo physics

The plasma-wave echo was experimentally reported by Malmberg, Wharton, Gould and O'Neil in 1968. A second perturbation couples to velocity-space structure prepared by the first and generates a difference mode whose phase later rephases. Echo amplitude depends on equilibrium distribution, pulse geometry, field-particle coupling, collisions, finite amplitude and measurement bandwidth. The phenomenon is established; no priority claim is made. [Malmberg et al. (1968), Ichikawa and Suzuki (1969), Grenier et al. (2022)]

4 Projection memory

For full state z(t), evolution dz/dt=R(z), and projection P onto resolved observables, with Q=I-P, Mori-Zwanzig identities express resolved dynamics through an instantaneous term, a memory integral and unresolved initial-state contributions. A finite lag window is an approximation, not the exact kernel. This release does not recover an exact kernel. [Zwanzig (1961), Mori (1965), Chorin et al. (2000), Burles et al. (2026)]

5 Two-pulse derivation

Let the first perturbation have wavenumber k1 at t=0 and the second have wavenumber k2 at t=tau. Coupling produces

$$k_3 = k_2 - k_1.$$

The velocity-dependent phase is proportional to k2*tau-k3*t. Setting it to zero gives

$$t_e = \frac{k_2 \tau}{k_2 - k_1}, \quad k_2 > k_1 > 0.$$

This is a P1 identity under the stated phase ansatz, not an amplitude theorem.

6 Reduced echo envelope

Define w=1/k3 and

$$E_3(t) = A_3 \exp[-(t - t_e)^2/(2w^2)], \quad A_3 = a_1 a_2 (k_1/k_3) \exp[-\nu k_1^2 \tau^3/3].$$

The bilinear pulse factor is a second-order ansatz; k1/k3 is a normalization choice; the Gaussian envelope is a smooth test signal. None is a universal physical coefficient.

7 Collisional attenuation

Velocity diffusion damps fine velocity-space Fourier structure at a rate proportional to the square of its velocity wavenumber. The cubic exponent is a transparent reduced proxy, not an exact Coulomb, BGK or Lenard-Bernstein echo law. No physical collision frequency is inferred from nu. [Ichikawa and Suzuki (1969)]

8 Synthetic memory benchmark

A deterministic synthetic signal combines a damped oscillation, a localized Gaussian echo-like pulse and Gaussian noise. Seed 20261110 is fixed. Examples are split chronologically, 70 percent for training and 30 percent for testing. Ridge regression with the current observation is compared with a 24-sample delay window. This is not a neural network and the target is not kinetic data.

9 Forecast result

Memoryless one-step RMSE is 0.0222991; 24-lag delay RMSE is 0.0160948, a 27.82 percent reduction on one synthetic holdout. This shows only that temporal context helps this constructed signal. It does not establish a physical memory kernel. A physical test needs independent kinetic trajectories and grouped holdouts.

10 Novelty and falsification

A finite-memory neural-operator closure for Landau damping appeared online on 30 October 2026. This work does not claim finite-memory plasma closure is new. A next-stage study must reproduce Landau damping, run a two-pulse echo, test grid convergence, monitor invariants, compare collision operators and hold out pulse timing, wavenumber, amplitude and collisionality.

11 Limitations

The reduced envelope is not a self-consistent Vlasov-Poisson solution. It omits field feedback, nonlinear trapping, physical collision operators, multidimensional geometry and instrument effects. Numerical dissipation can mimic collisional attenuation. The synthetic predictor may exploit autocorrelation rather than physical memory. No laboratory or MMS data are analysed.

12 Candidate table

Candidate	Decision	Reason
Neural Landau closure	Reject	Direct precedent
Echo memory test	Select	Falsifiable intervention
Hermite inversion	Defer	Needs kinetic solver
Collision identification	Defer	Needs validated operators
MMS echo	Defer	No controlled pulse

13 Assumption table

Assumption	Role	Boundary
1D kinematics	Phase cancellation	Not multidimensional
Free streaming	Echo geometry	No field feedback
Second-order pulse	Bilinear ansatz	Small-signal only
Gaussian envelope	Smooth signal	Not universal
Cubic attenuation	Diffusion proxy	Operator dependent

14 Verification table

Check	Result	Meaning
Echo time	6.0	Identity check
Amplitude ratio	2.0	Model property
Attenuation	Monotone	Declared formula
Proxy sum	1.0	Normalization
Invalid k	Rejected	Input guard
Delay RMSE	27.82 percent lower	Synthetic only

15 Literature table

Work	Contribution	Relation
Malmberg 1968	Echo experiment	Known physics
Ichikawa 1969	Collisions	Operator effects
Grenier 2022	Echo cascades	Nonlinear context
Zwanzig/Mori	Projection memory	Foundations
Burles 2026	Neural closure	Adjacent prior

16 Falsification table

Claim	Test	Failure
Echo time	Kinetic holdouts	Timing error
Memory gain	History shuffle	Gain vanishes
Collision law	Operator comparison	Resolution dependence
Generalization	Pulse holdouts	OOD collapse
Invariants	Residual checks	Excess drift

17 Reproducible figures

Research-process disclosure

Co-equal research-process credit is given to Devanik Debnath and Xylia, the AI-assisted research orchestration system coordinating discovery, synthesis and verification. Devanik Debnath retains responsibility for the final manuscript, scientific claims, citations and released artifacts.

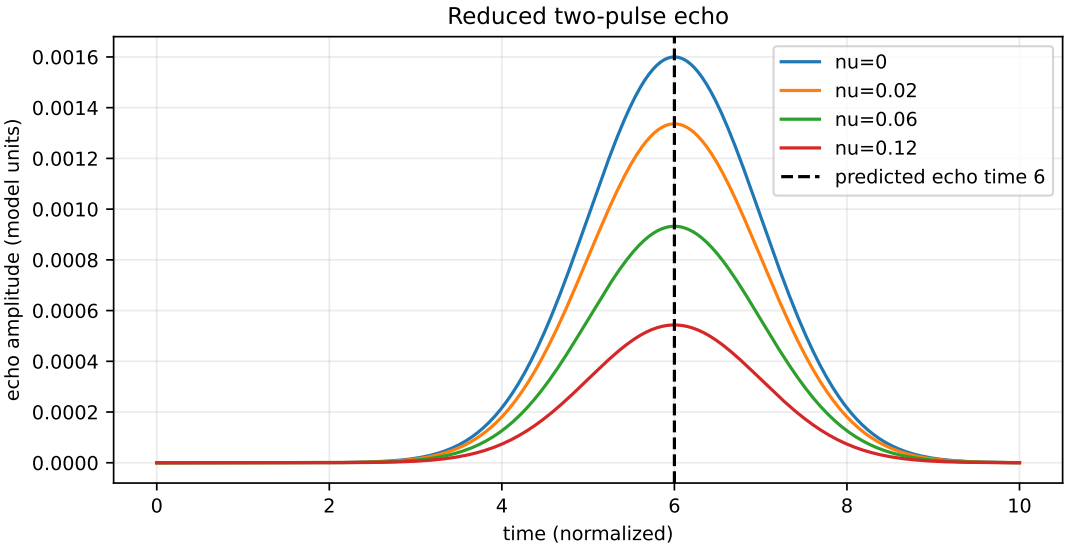


Figure 1: Reduced echo envelope.

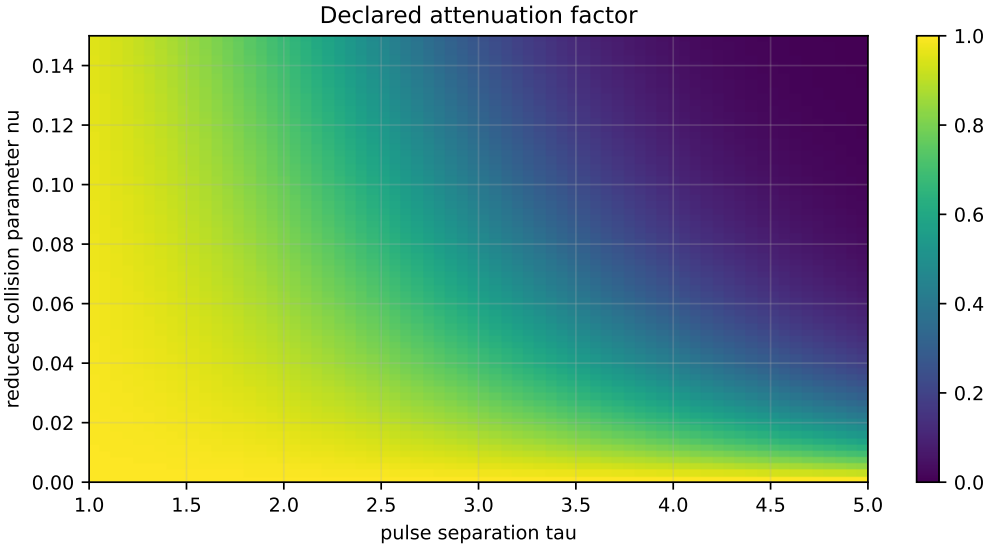


Figure 2: Attenuation map.

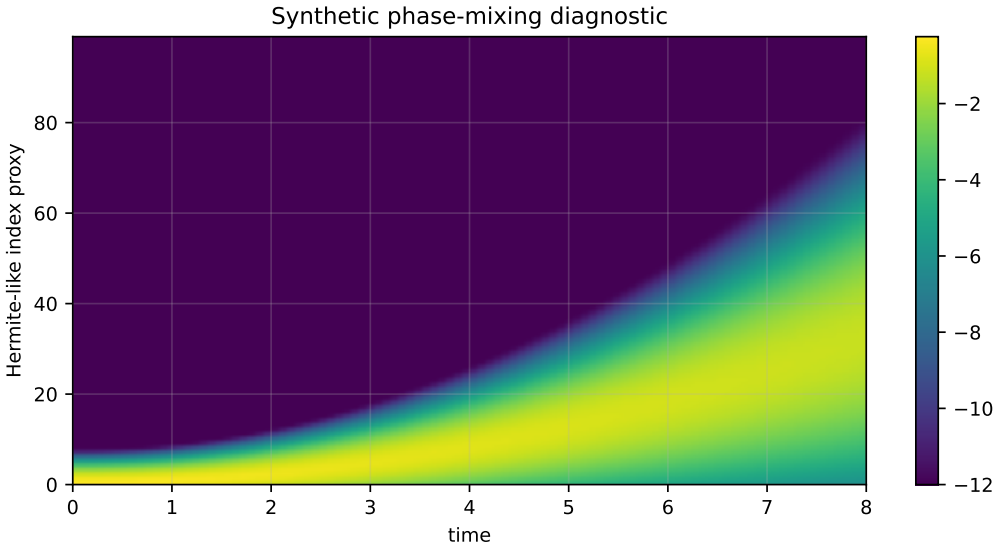


Figure 3: Synthetic phase-mixing proxy.

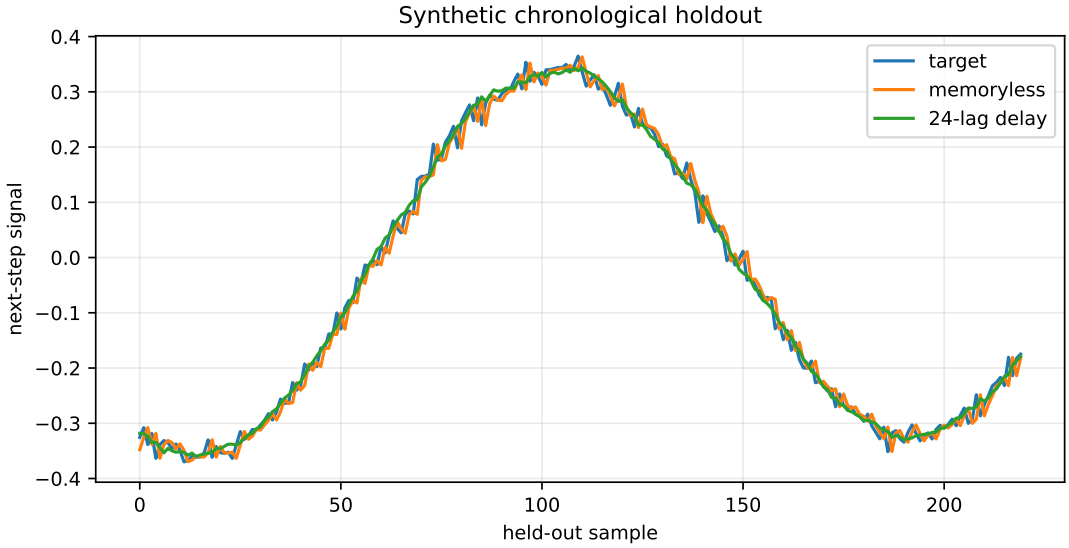


Figure 4: Synthetic memory forecast.

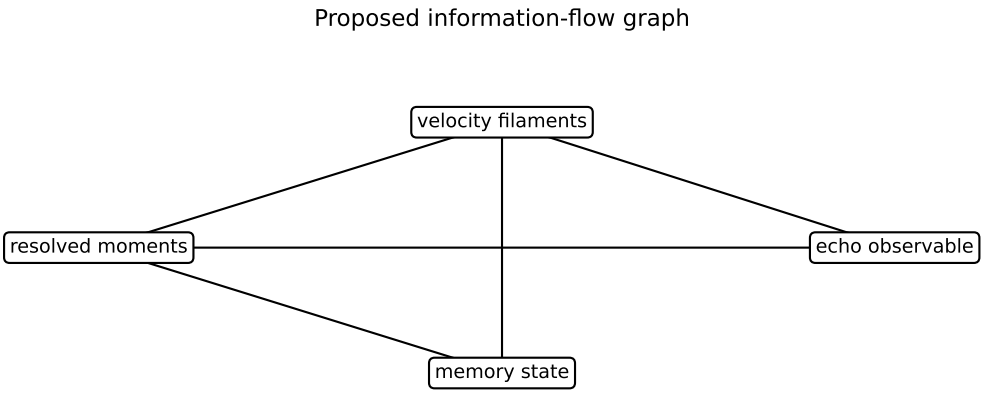


Figure 5: Information-flow graph.

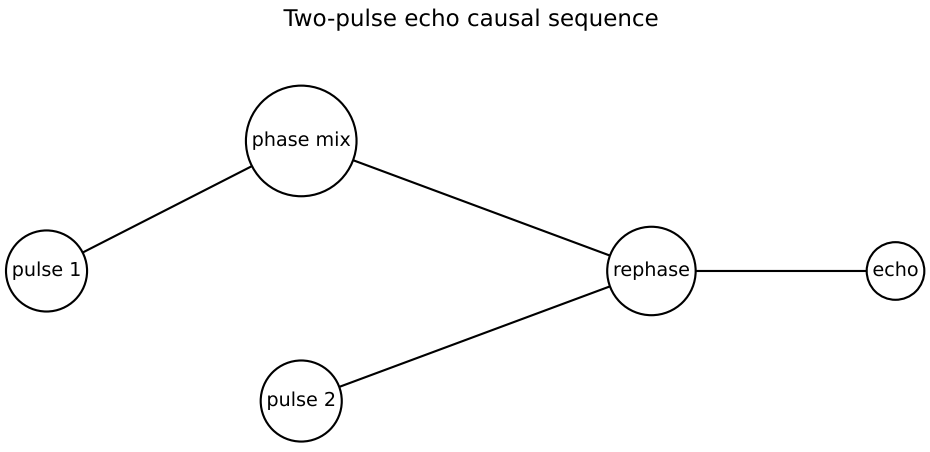


Figure 6: Echo causal graph.

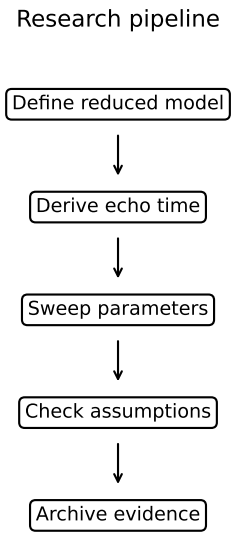


Figure 7: Research pipeline.

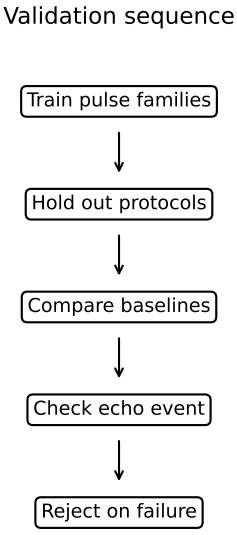


Figure 8: Validation flow.

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